$$
\begin{array}{r|cc|}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0 \\
\hline
\end{array}
$$

$$
\begin{array}{r|cccc|}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \mathrm{Y}_1 k^2 & \frac{\mathrm{Y}_1 k^2}{\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{\mathrm{Y}_1 k^2}{\sqrt{2}} & \frac{\mathrm{Y}_1 k^2}{2} & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{\mathrm{Y}_2 k^2}{2} & -\frac{\mathrm{Y}_3 k^2}{2\sqrt{2}} \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & -\frac{\mathrm{Y}_3 k^2}{2\sqrt{2}} & \frac{\mathrm{Y}_4 k^2}{2} \\
\hline
\end{array}
$$

$$
\begin{array}{r|cc|}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{3 k^2 \overset{(4)}{\kappa}_{15}}{2} \\
\hline
\end{array}
$$

$$
\begin{array}{r|cc|}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0 \\
\hline
\end{array}
$$

$$
\begin{array}{r|cccc|}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{2}{3\,\mathrm{Det}(1^+)} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & \frac{1}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{\mathrm{Y}_4 k^2}{2\,\mathrm{Det}(1^-)} & \frac{\mathrm{Y}_3 k^2}{2\sqrt{2}\,\mathrm{Det}(1^-)} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{\mathrm{Y}_3 k^2}{2\sqrt{2}\,\mathrm{Det}(1^-)} & \frac{\mathrm{Y}_2 k^2}{2\,\mathrm{Det}(1^-)} \\
\hline
\end{array}
$$

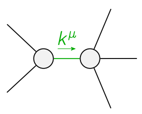
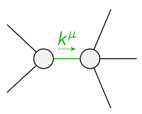
$$
\begin{array}{r|cc|}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{2}{3 k^2 \overset{(4)}{\kappa}_{15}} \\
\hline
\end{array}
$$

Abbreviations used in matrices

$$\mathrm{Y}_1 == 2\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{3} \;\&\&\; \mathrm{Y}_2 == 2\overset{(4)}{\kappa}_{1} + 3\overset{(4)}{\kappa}_{15} \;\&\&$$
$$\mathrm{Y}_3 == 2\overset{(4)}{\kappa}_{1} - \overset{(4)}{\kappa}_{7} \;\&\&\; \mathrm{Y}_4 == \overset{(4)}{\kappa}_{1} - 3\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{7} \;\&\&\; \mathrm{Det}(1^+) == \frac{3}{2} k^2 (2\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{3})$$

Lagrangian

$$\overset{(4)}{\kappa}_{1}\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} - \overset{(4)}{\kappa}_{1}\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_{3}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} +$$
$$6\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - 2\overset{(4)}{\kappa}_{3}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} + 3\overset{(4)}{\kappa}_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \overset{(4)}{\kappa}_{3}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} +$$
$$\overset{(4)}{\kappa}_{7}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \frac{1}{2}\overset{(4)}{\kappa}_{3}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}$$

<table>
<tr><td>Added source term(s):</td><td colspan="3">$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$</td></tr>
<tr><td>Source constraint(s)</td><td># constraint(s)</td><td colspan="2">Covariant form</td></tr>
<tr><td>$\mathcal{J}^{\#1}_{0^-}{}^{\alpha\beta\chi} == 0$</td><td>1</td><td colspan="2">$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} ==$
$\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$</td></tr>
<tr><td>$\mathcal{J}^{\#1}_{0^+} == 0$</td><td>1</td><td colspan="2">$\partial_\beta\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$</td></tr>
<tr><td>$\mathcal{J}^{\#1}_{1^+}{}^{\alpha\beta} == \mathcal{J}^{\#2}_{1^+}{}^{\alpha\beta}$</td><td>3</td><td colspan="2">$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} + 2\,\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$</td></tr>
<tr><td>$\mathcal{J}^{\#1}_{2^+}{}^{\alpha\beta} == 0$</td><td>5</td><td colspan="2">$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} + 3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$</td></tr>
<tr><td>Total # constraint(s):</td><td>10</td><td colspan="2"></td></tr>
<tr><td>Resolved pole(s)</td><td># polarization(s)</td><td>Square mass</td><td>Residue</td></tr>
<tr><td></td><td>2</td><td>0</td><td>$(18\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}{}^{2} + 18\overset{(4)}{\kappa}_{15}{}^{3} + 18\overset{(4)}{\kappa}_{15}{}^{2}\overset{(4)}{\kappa}_{3} + 6\overset{(4)}{\kappa}_{15}{}^{2}\overset{(4)}{\kappa}_{7} + 6\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{7} - \overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{7}{}^{2} + 2\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{7}{}^{2} -$
$\sqrt{(324\overset{(4)}{\kappa}_{1}{}^{2}\overset{(4)}{\kappa}_{15}{}^{4} + 8100\overset{(4)}{\kappa}_{15}{}^{6} - 1080\overset{(4)}{\kappa}_{15}{}^{5}(3\overset{(4)}{\kappa}_{3} - 5\overset{(4)}{\kappa}_{7}) + 2\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3}(12\overset{(4)}{\kappa}_{3} - 11\overset{(4)}{\kappa}_{7})\overset{(4)}{\kappa}_{7}{}^{3} + 4\overset{(4)}{\kappa}_{3}{}^{2}\overset{(4)}{\kappa}_{7}{}^{4} +}$
$36\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}{}^{2}(5\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{3})(18\overset{(4)}{\kappa}_{15}{}^{2} + 6\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{7} + \overset{(4)}{\kappa}_{7}{}^{2}) + 108\overset{(4)}{\kappa}_{15}{}^{4}(3\overset{(4)}{\kappa}_{3}{}^{2} - 20\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{7} + 18\overset{(4)}{\kappa}_{7}{}^{2}) +$
$12\overset{(4)}{\kappa}_{15}{}^{3}\overset{(4)}{\kappa}_{7}(18\overset{(4)}{\kappa}_{3}{}^{2} - 72\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{7} + 29\overset{(4)}{\kappa}_{7}{}^{2}) + \overset{(4)}{\kappa}_{15}{}^{2}\overset{(4)}{\kappa}_{7}{}^{2}(108\overset{(4)}{\kappa}_{3}{}^{2} - 168\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{7} + 37\overset{(4)}{\kappa}_{7}{}^{2})))/$
$(\overset{(4)}{\kappa}_{15}(2\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{3})(6\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15} + 18\overset{(4)}{\kappa}_{15}{}^{2} + 6\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{7} + \overset{(4)}{\kappa}_{7}{}^{2}))$</td></tr>
<tr><td></td><td>2</td><td>0</td><td>$(18\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}{}^{2} + 18\overset{(4)}{\kappa}_{15}{}^{3} + 18\overset{(4)}{\kappa}_{15}{}^{2}\overset{(4)}{\kappa}_{3} + 6\overset{(4)}{\kappa}_{15}{}^{2}\overset{(4)}{\kappa}_{7} + 6\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{7} - \overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{7}{}^{2} + 2\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{7}{}^{2} +$
$\sqrt{(324\overset{(4)}{\kappa}_{1}{}^{2}\overset{(4)}{\kappa}_{15}{}^{4} + 8100\overset{(4)}{\kappa}_{15}{}^{6} - 1080\overset{(4)}{\kappa}_{15}{}^{5}(3\overset{(4)}{\kappa}_{3} - 5\overset{(4)}{\kappa}_{7}) + 2\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{3}(12\overset{(4)}{\kappa}_{3} - 11\overset{(4)}{\kappa}_{7})\overset{(4)}{\kappa}_{7}{}^{3} + 4\overset{(4)}{\kappa}_{3}{}^{2}\overset{(4)}{\kappa}_{7}{}^{4} +}$
$36\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15}{}^{2}(5\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{3})(18\overset{(4)}{\kappa}_{15}{}^{2} + 6\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{7} + \overset{(4)}{\kappa}_{7}{}^{2}) + 108\overset{(4)}{\kappa}_{15}{}^{4}(3\overset{(4)}{\kappa}_{3}{}^{2} - 20\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{7} + 18\overset{(4)}{\kappa}_{7}{}^{2}) +$
$12\overset{(4)}{\kappa}_{15}{}^{3}\overset{(4)}{\kappa}_{7}(18\overset{(4)}{\kappa}_{3}{}^{2} - 72\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{7} + 29\overset{(4)}{\kappa}_{7}{}^{2}) + \overset{(4)}{\kappa}_{15}{}^{2}\overset{(4)}{\kappa}_{7}{}^{2}(108\overset{(4)}{\kappa}_{3}{}^{2} - 168\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{7} + 37\overset{(4)}{\kappa}_{7}{}^{2})))/$
$(\overset{(4)}{\kappa}_{15}(2\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{3})(6\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{15} + 18\overset{(4)}{\kappa}_{15}{}^{2} + 6\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{7} + \overset{(4)}{\kappa}_{7}{}^{2}))$</td></tr>
<tr><td colspan="2">Resolved unitarity condition(s):</td><td colspan="2">$(\overset{(4)}{\kappa}_{15} < 0 \;\&\&\; \overset{(4)}{\kappa}_{3} < 2\overset{(4)}{\kappa}_{15} \;\&\&\; \overset{(4)}{\kappa}_{1} < \frac{-18\overset{(4)}{\kappa}_{15}{}^{2} - 6\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{7} - \overset{(4)}{\kappa}_{7}{}^{2}}{6\overset{(4)}{\kappa}_{15}}) \;\|$
$(\overset{(4)}{\kappa}_{15} > 0 \;\&\&\; \overset{(4)}{\kappa}_{3} < 2\overset{(4)}{\kappa}_{15} \;\&\&\; \overset{(4)}{\kappa}_{1} < \frac{-18\overset{(4)}{\kappa}_{15}{}^{2} - 6\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{7} - \overset{(4)}{\kappa}_{7}{}^{2}}{6\overset{(4)}{\kappa}_{15}})$</td></tr>
</table>